

Structured Agentic Memory Access for Conversational AI

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TL;DR: Semantic XPath is a token-efficient and high-performing alternative to RAG for conversational agent using trees to semantically structure conversation history and semantic extensions of XPath queries to manage memory access.**1 Conversational Memory**

A long-term conversational agent must read, update, and revise an evolving conversation history to generate self-coherent responses aligned with the user's request.

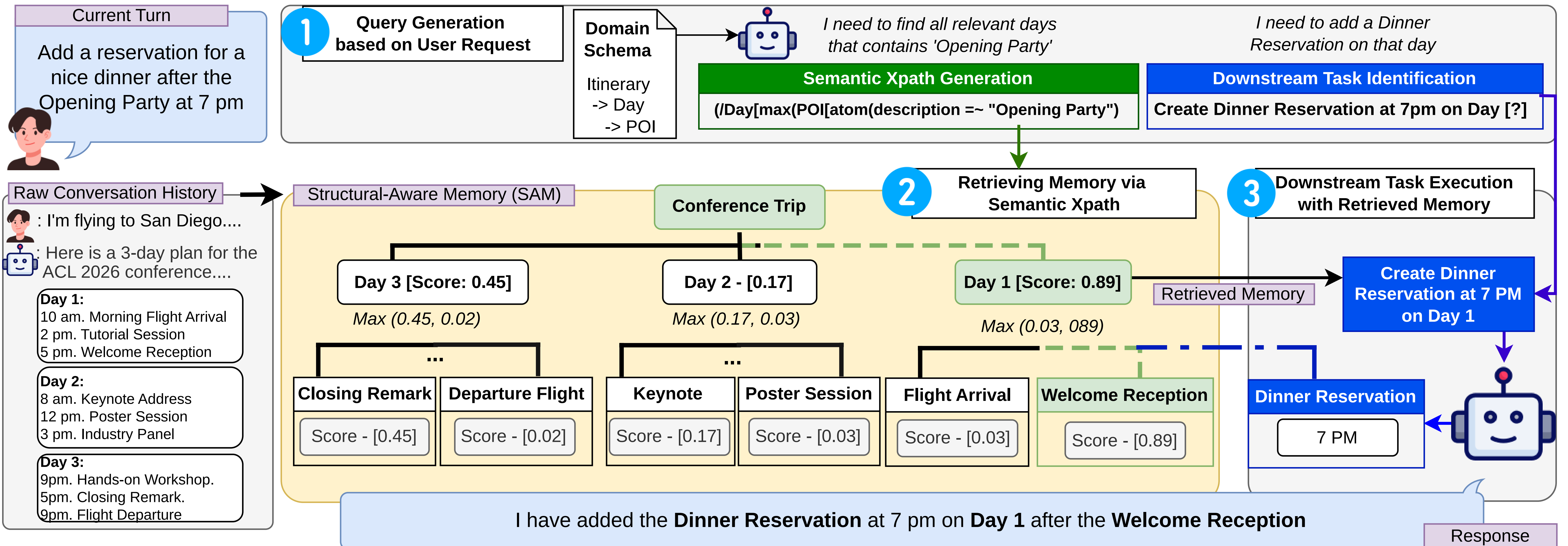
- In-context memory** is limited by context length and incurs high token costs.
- Flat RAG** relies on dense retrieval, which often struggles to target specific memories from long conversational histories.

Problem: current memory access either has a high tokens cost or fail to utilize the possible internal structure of the conversational history

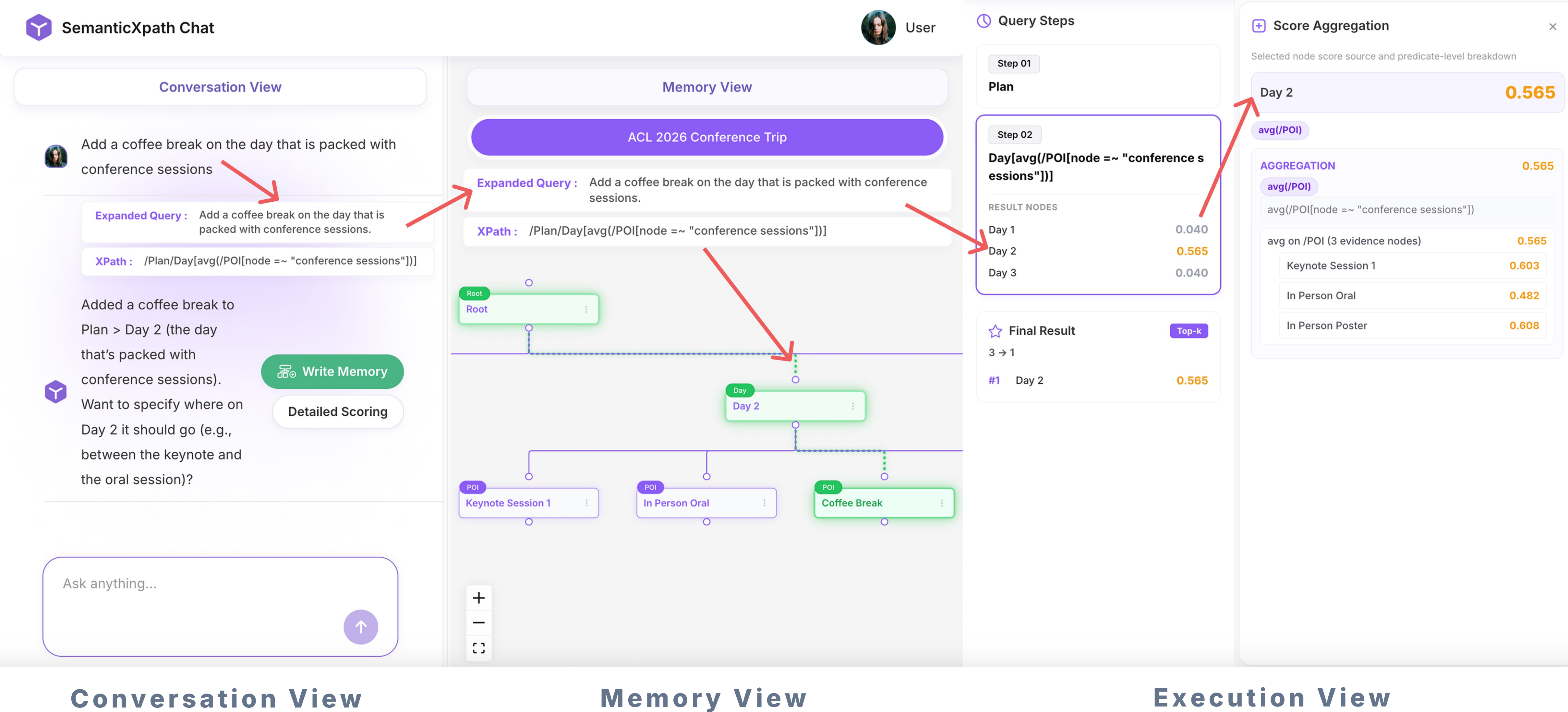
We propose **Semantic XPath**: a structured memory approach that organizes conversational history as trees and generates XPath-style queries with semantic matching, enabling precise and targeted access to relevant memories.

2 What is Semantic XPath?

an XPath-style navigation mechanism for accessing and reasoning over structured memory

**3 Introducing Semantic Xpath Chat**

A ready-to-use, semantically structured-aware long-term conversational agent built on Semantic XPath.

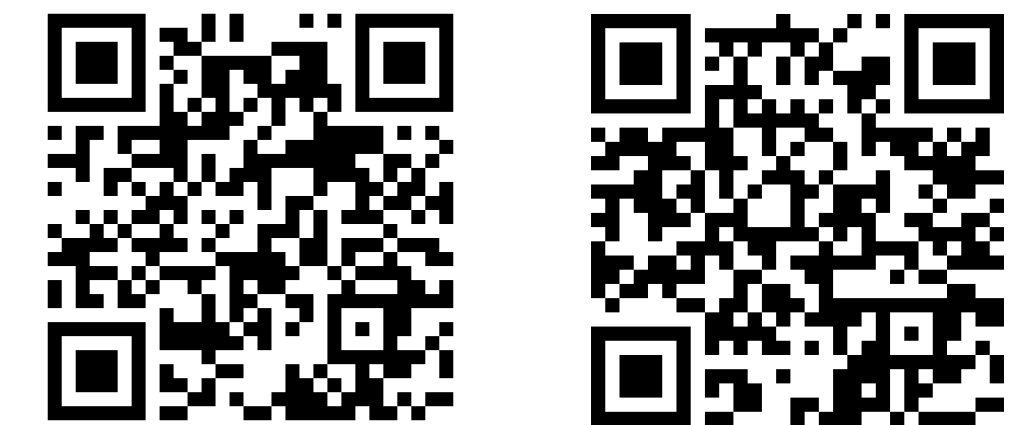


Conversation View

Memory View

Execution View

- Conversation View** displays the dialogue history and lets users jump from each turn to the corresponding memory and execution details.
- Memory View** visualizes the tree-structured memory, highlighting the relevant path and substructure for the current user request.
- Execution View** shows step-by-step query execution and semantic scoring, enabling users to inspect candidate nodes and relevance scores.

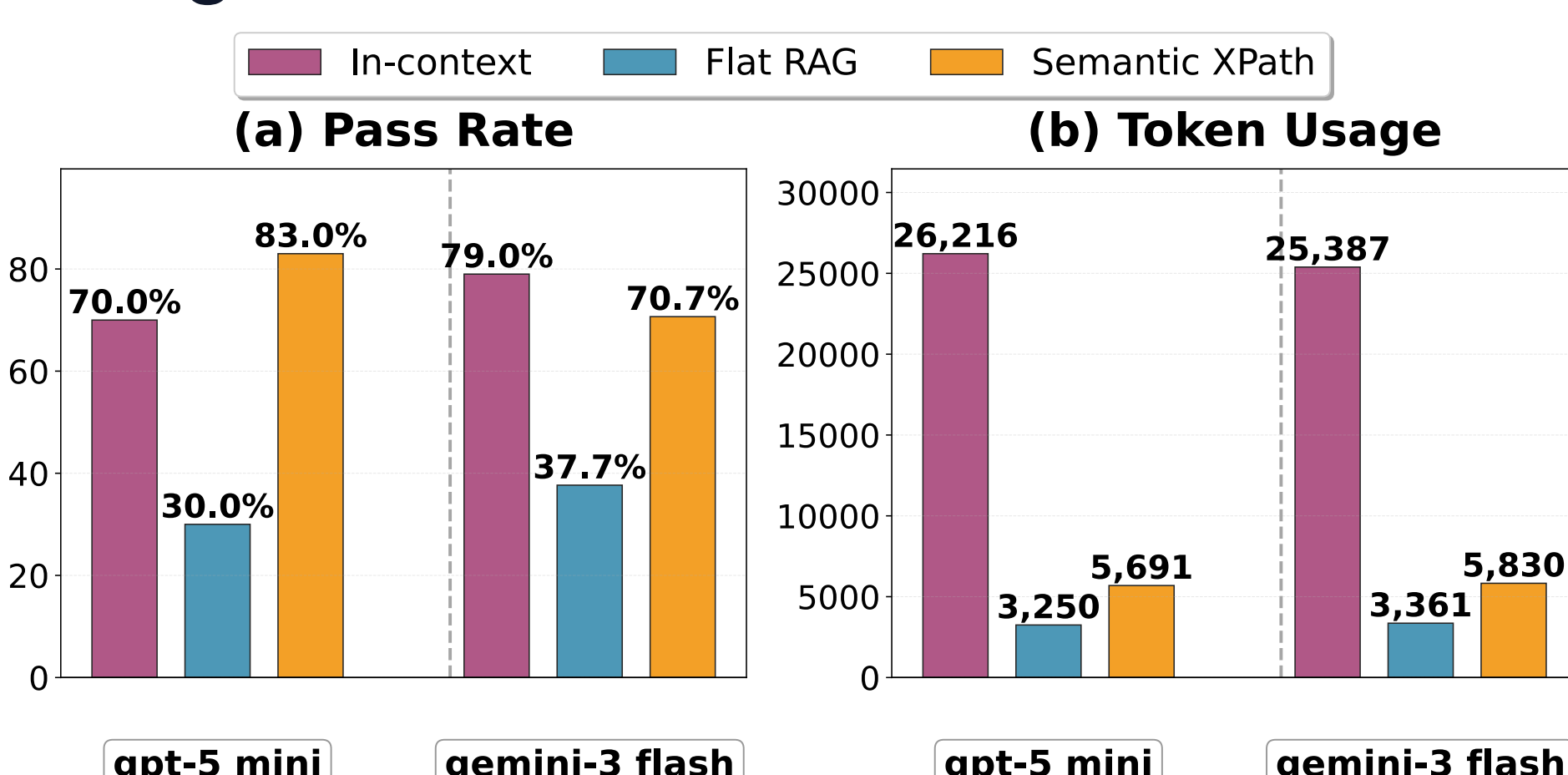
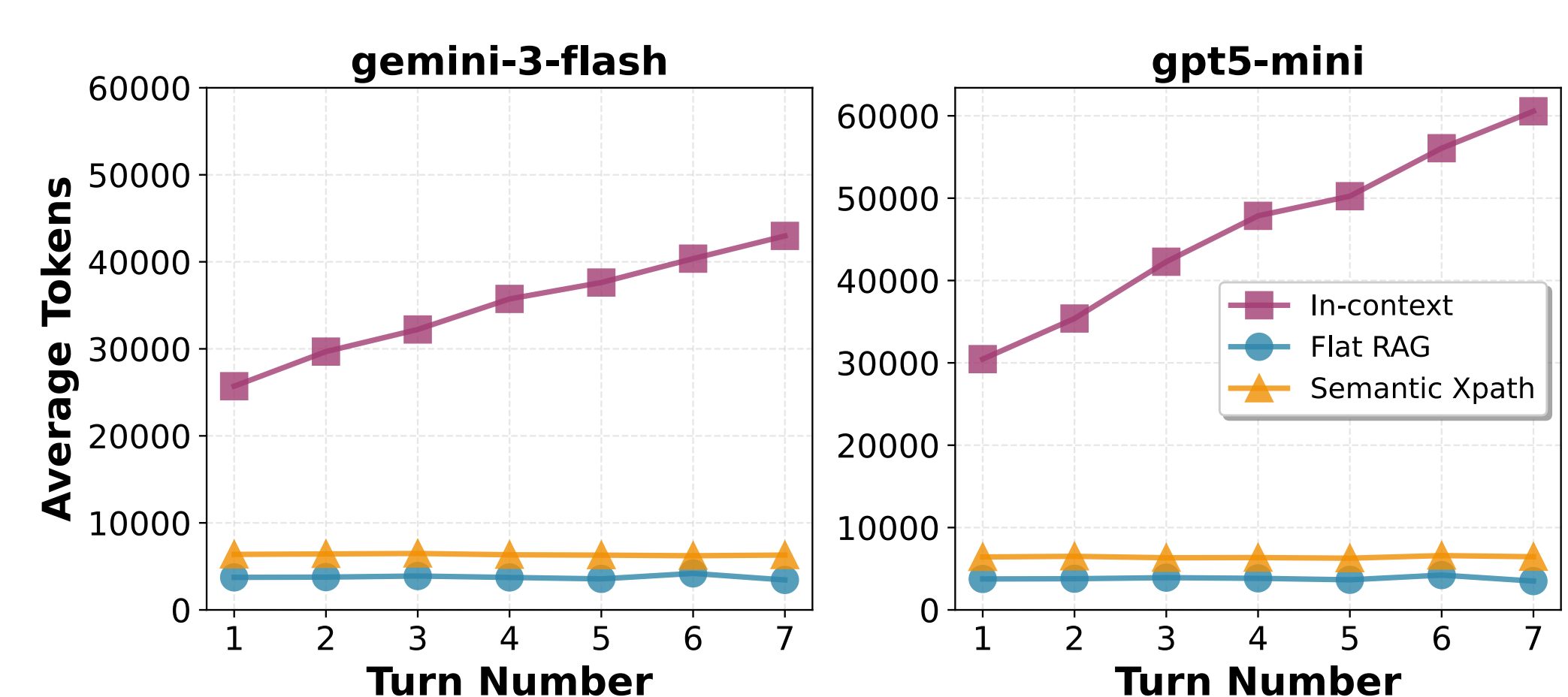
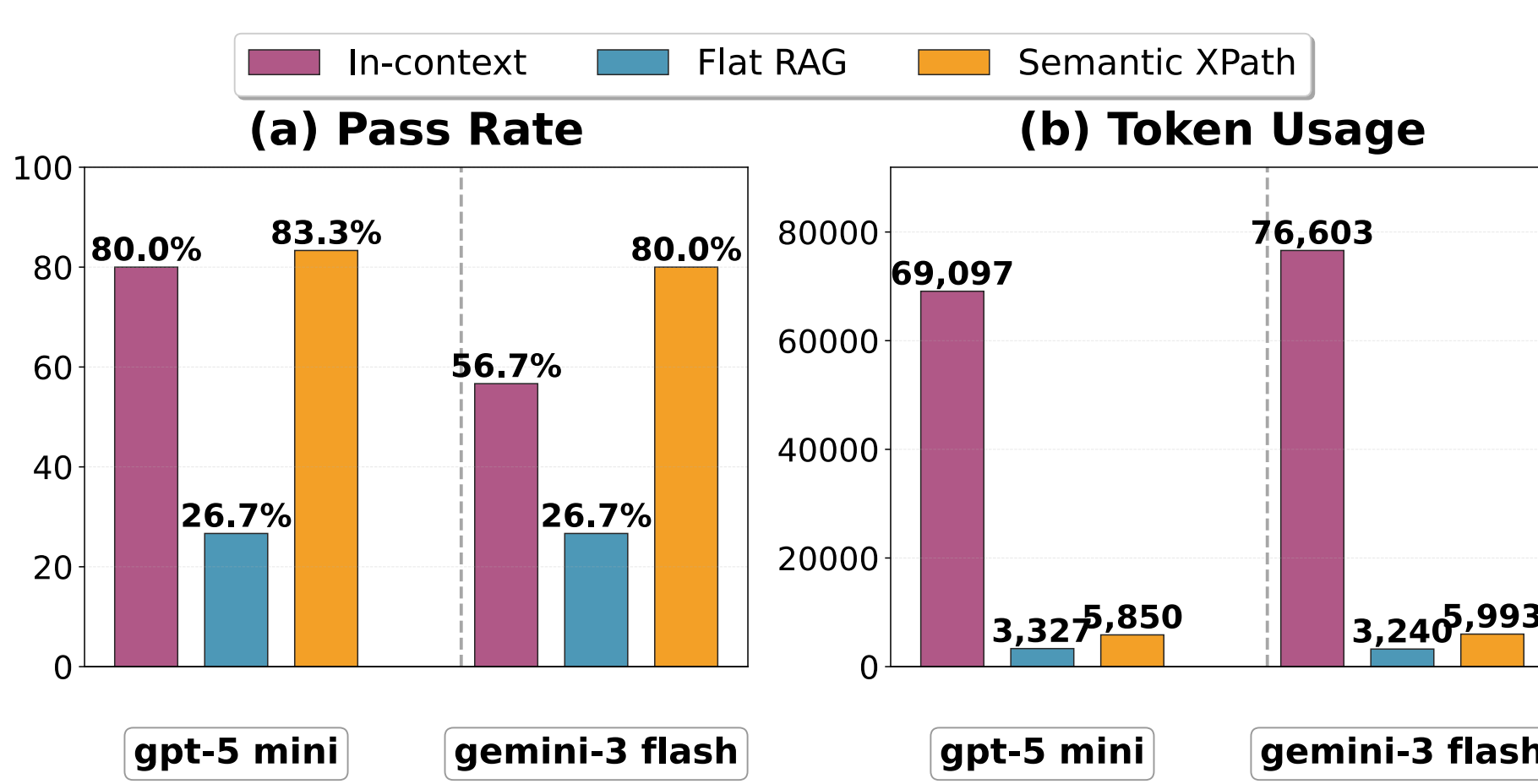


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4 Does it work?

- Benchmarks:** 3 curated domains
- Models:** GPT-5 mini, Gemini 3 Flash

- Baselines:** in-context memory, Flat RAG
- Metrics:** Pass Rate / answer score, input tokens

Single Turn tasks**Multi-turn tasks**

Stable performance with far lower token usages.

In Context methods token usage increase significantly as turn increase.

